Assignment 7.2

Week 7

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AIT620

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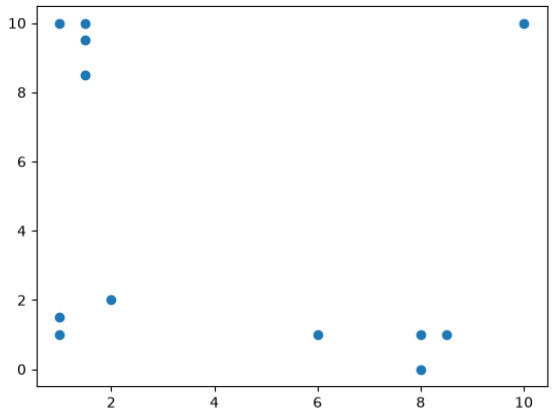
Code

JupyterLab Python 3 (ipykernel)

[2]:
```
import numpy as np
data_points = np.array([[1, 1], [1, 1.5], [2, 2], \
[8, 1], [8, 0], [8.5, 1], \
[6, 1], [1, 10], [1.5, 10], \
[1.5, 9.5], [10, 10], [1.5, 8.5]])
```

[3]:
```
import matplotlib.pyplot as plot
plot.scatter(data_points.transpose()[0], \
data_points.transpose()[1])
```

[3]:
```
<matplotlib.collections.PathCollection at 0x154e552b0>
```

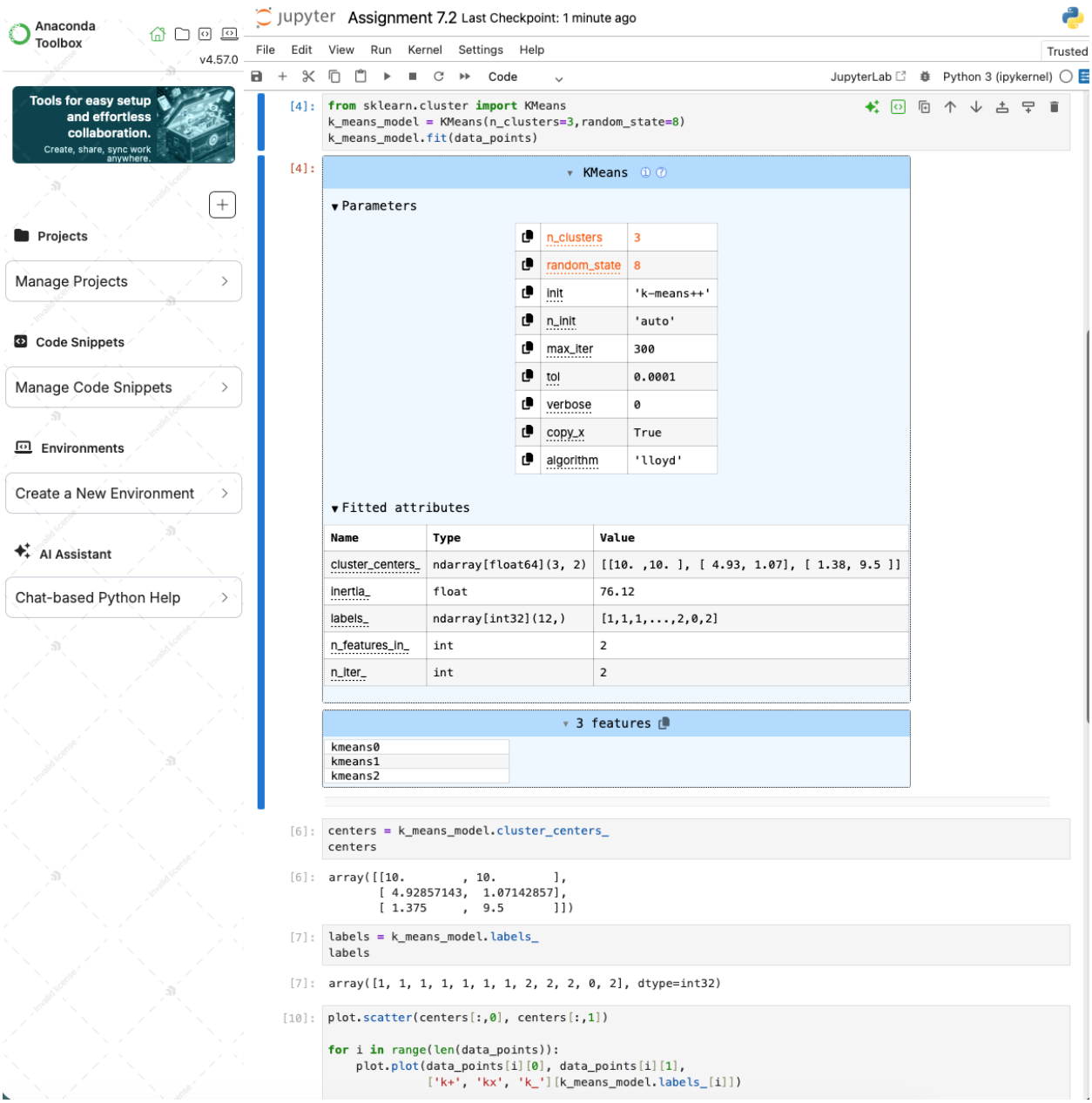


[4]:
```
from sklearn.cluster import KMeans
k_means_model = KMeans(n_clusters=3,random_state=8)
k_means_model.fit(data_points)
```

[4]: KMeans

Parameters

n_clusters	3
random_state	8
init	'k-means++'
n_init	'auto'
max_iter	300
tol	0.0001
verbose	0
copy_x	True
algorithm	'lloyd'


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Code
JupyterLab
Python 3 (ipykernel)
[4]: from sklearn.cluster import KMeans
k_means_model = KMeans(n_clusters=3,random_state=8)
k_means_model.fit(data_points)
KMeans
Parameters
n_clusters 3
random_state 8
init 'k-means++'
n_init 'auto'
max_iter 300
tol 0.0001
verbose 0
copy_x True
algorithm 'lloyd'
Fitted attributes
Name Type Value
cluster_centers_ ndarray[float64](3, 2) [[10. ,10.], [4.93, 1.07], [1.38, 9.5]]
inertia_ float 76.12
labels_ ndarray[int32](12,) [1,1,1,...,2,0,2]
n_features_in_ int 2
n_iter_ int 2
3 features
kmeans0
kmeans1
kmeans2
[6]: centers = k_means_model.cluster_centers_
centers
[6]: array([[10. , 10.],
[4.92857143, 1.07142857],
[1.375 , 9.5]])
[7]: labels = k_means_model.labels_
labels
[7]: array([1, 1, 1, 1, 1, 1, 2, 2, 2, 0, 2], dtype=int32)
[10]: plot.scatter(centers[:,0], centers[:,1])
for i in range(len(data_points)):
plot.plot(data_points[i][0], data_points[i][1],
['k+', 'kx', 'k_'][k_means_model.labels_[i]])

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Code

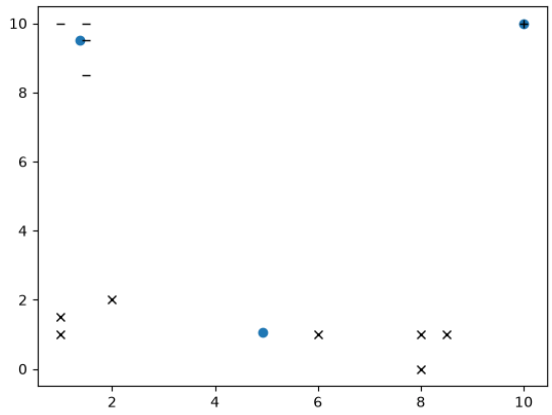
JupyterLab

Python 3 (ipykernel)

[10]:
```
plot.scatter(centers[:,0], centers[:,1])

for i in range(len(data_points)):
    plot.plot(data_points[i][0], data_points[i][1],
              ['k+', 'kx', 'k_'][k_means_model.labels_[i]])

plot.show()
```



[12]:
```
k_means_model = KMeans(n_clusters=2, random_state=8)
k_means_model.fit(data_points)

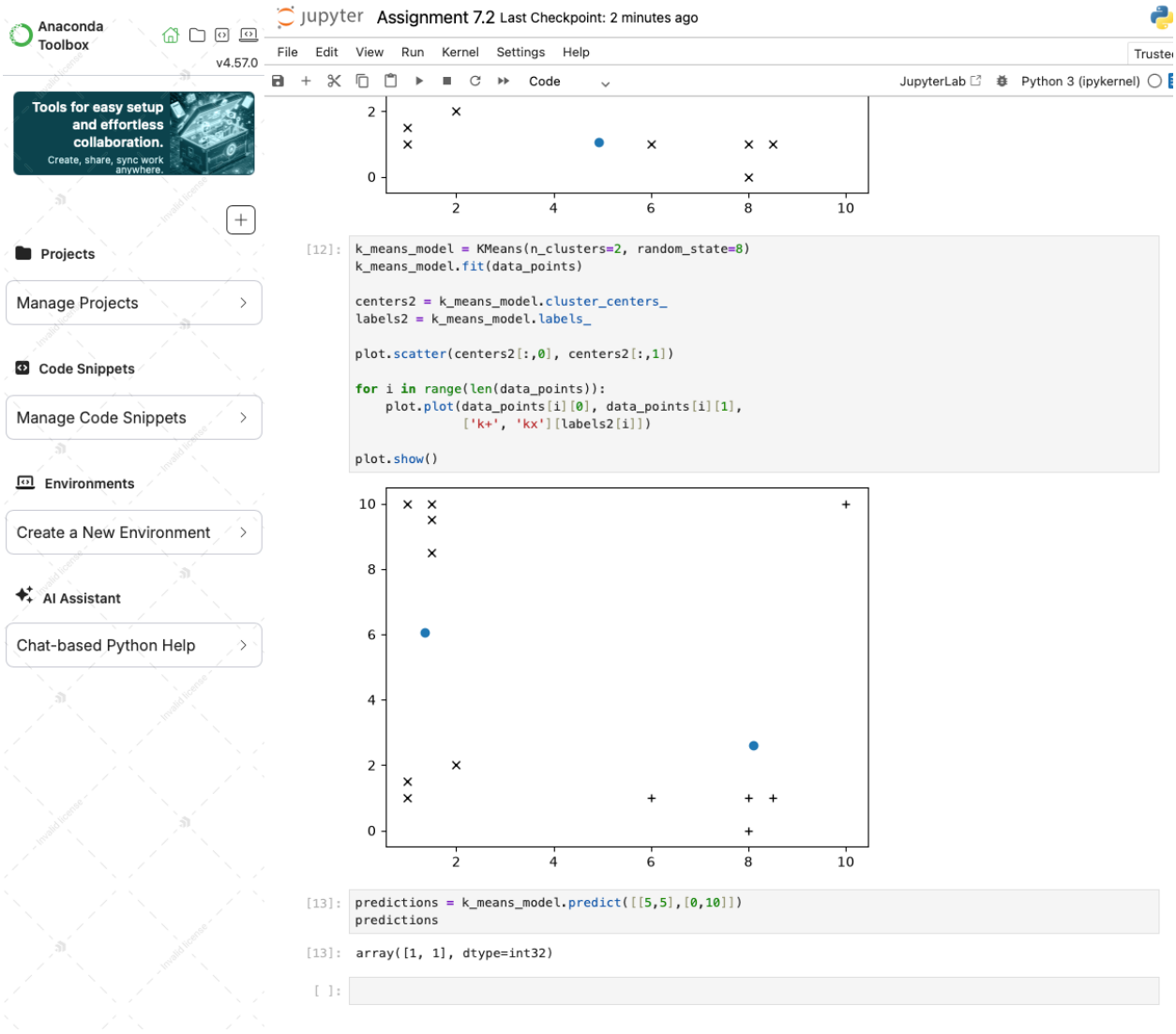
centers2 = k_means_model.cluster_centers_
labels2 = k_means_model.labels_

plot.scatter(centers2[:,0], centers2[:,1])

for i in range(len(data_points)):
    plot.plot(data_points[i][0], data_points[i][1],
              ['k+', 'kx'][labels2[i]])

plot.show()
```





Conclusion:

The data points were clustered using K-Means clustering algorithm in this assignment to group the data points depending on their similarity.

To look for trends in the data, initially a scatter plot was created. Three clusters were used for the K-Means algorithm. We found three different centers of the clusters in the results.

Then, the algorithm was run again with two clusters. Differences between the two models were shown by the scatter plots. Every point of the data set was matched with the clusters according to their distances from the cluster means. Clusters were also predicted for new data points using the model.

This showed how increasing the number of clusters or decreasing the number of clusters would impact the grouping.Overall, this activity demonstrated the ability of K-Means to find patterns in 'noisy' data without labels.